Curriculum Vitae

Name- Tirtha Das Banerjee
Gender - Male
Permanent Residency- Singapore (SPR)
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Language- English
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Google Scholar: https://scholar.google.com/citations?user=H2S5ROQAAAAJ&hl=en
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Orchid: https://orcid.org/0000-0002-6092-0824
GitHub: https://github.com/tdblab
h-index: 12
i10-index: 13

Current Positions

Senior Postdoctoral Research Fellow: Department of Biological Sciences, National University of Singapore.
Roles: 1) Integration of single cell, LCM-seq, and spatial mRNA technologies to answer the development and evolution of serial homologs.
2) Exploring the combined role of reaction-diffusion and positional information in venation patterning.
3) Identify the key Wnt signaling network genes in pattern formation during different stages of insect wing development.
4) Development of CORN-DRIVE system and insect colony optimization for insectary (Evo-Devo Lab).
3rd Jan 2025-Present

Research Associate: Yong Loo Lin School of Medicine, National University of Singapore.
Roles: 1) Development of automation systems (FISHER) for neurobehavioral assays.
2) Implementation of next generation multiplex mRNA profiling technology.
3)Development of user friendly RNAseq pipeline for implementation if zebrafish and killifish models.
4) Development of a portable spectroscope for detection of cortisol levels in aquaculture.
19th May 2025-Present

Past Positions

Academic

Senior Postdoctoral research Fellow: Yong Loo Lin School of Medicine and Department of Biological Sciences, National University of Singapore.
Project: Development and application of next generation spatial transcriptomics technologies.
chrna3 modulates alcohol response in zebrafish.
Date: June 2024-January 2025

Postdoctoral research Fellow: Department of Biological Sciences, National University of Singapore.
Projects: 1) Evolution and Development of novel complex traits.
2) Development and application of spatial transcriptomics technologies (in collaboration with GIS-Singapore).
Date: November 2020-June 2024

Research Scholar: Department of Civil Engineering, Curtin University, Perth, Western Australia.
Project: Development of self-healing concrete using cyanobacteria.
Date: January 2016-July 2016

Industrial

Orchid Chemicals and Pharmaceuticals Ltd., R&D Center, Sholinganallur, Chennai, India (acquired by Pfizer).
Field of research: Large scale bacteria-based enzyme production in 200-500L bioreactors and recombinant DNA technology.
Date: August 2015-June 2016

PhD Thesis

Molecular Mechanisms Underlying Venation and Color Patterning in *Bicyclus anynana* Butterflies. (2020). https://scholarbank.nus.edu.sg/handle/10635/186943

Supervisor: Prof. Antónia Monteiro

Educational Qualification

	Degree	Year	Institute	CAP/%
1	Doctor of Philosophy (Biological Sciences)	2016-2021	National University of Singapore	4.06/5
2.	Bachelor of Technology (Biotechnology)	2011-2015	National Institute of Technology, Durgapur	8.7/10*
3.	12th (CBSE Board)	2008-2010	Kendriya Vidyalaya, C.R.P.F., Durgapur, India	86.6%**
4.	10th (CBSE Board)	2007-2008	Kendriya Vidyalaya, C.R.P.F., Durgapur, India	83.8%

* First Class with Distinction.

**School Topper in Science Stream.

Biological Skills and expertise

Single cell and bulk RNA-Seq, multiplex-FISH, colony optimization and automation, spatial biology, laser capture microdissection, CRISPR-Cas9, immunohistochemistry, qPCR, gene cloning, western blotting, microbial gene and protein expression, and advanced bioinformatics.

Area of Interest

Molecular Biology, Developmental Biology, Evolutionary Biology, Insect Biology, Bioengineering, Robotics, Microbiology, Artificial Intelligence, Computer Vision, and Spatial Biology.

Model organisms

Bicyclus anynana and *Danio rerio*

Programming languages I use

C++, html, R, and Python

Hardware and Microcontrollers

Arduino (ATMega2560, ATMega328p), Esp32, AMB82-mini, STM32

Awards and Scholarships

Inspiring Research Mentor Award, 2025. Awarded by NUS High School of Math and Science.
Young Researcher Award, Asia Evo-Devo-2021. Awarded by Asia Evo-devo 2021.
Yale-NUS scholarship (2016-2020). Awarded by Yale University and NUS.

Grants Awarded

1) Integrating Reaction-Diffusion and Positional-Information in patterning the wing veins of *Bicyclus anynana* butterflies. (MOE Tier 1, primarily written, PI: Antónia Monteiro).
2) The evolution of novel complex traits from preexisting traits: Identifying shared cellular identities and gene regulation across traits. (MOE Tier 2, co-written with Antónia Monteiro and Suriya Narayanan Murugeshan).
3) Investigating a transcriptional role of primary cilia in striatal neural computation mediating cognitive flexibility. (MOE Tier 2, as collaborator, PI: Siew Cheng Phua).

Technologies developed

1) RAM-FISH: Spatial mRNA profiling technology that allows multiplexing (~30 genes) in complex biological systems such as whole mount organs of butterflies and zebrafish. Current institutions using the systems: Department of Biological Sciences and Yong Loo Lin School of Medicine, NUS; and Institute of Cellular and Molecular Biology, A*STAR.
2) NanoPulse: A nanolitre scale microinjection system for CRISPR-Cas9 and embryonic research application. Developed in collaboration with JayaVardhan Harihar Prasad and Chan Sum Ngai Jeff from Raffles Institution, Singapore. Awarded Silver at the 26th Science and Engineering Fair, Singapore - 2026.
3) FISHER: Flexible Instrument for Stimulus-controlled Habituation and Experimental Reinforcement for neurobehavioral assays, associative learning, and drug delivery. Four variants: FISHER_mini (mini version of FISHER for tight spaces), FISHER_delta (for high and low standard diet), and FISHER_Cam (a camera powered system to perform recordings while the experiments are being conducted).

Technologies currently under development

4) CORN-DRIVE: A multi-sensor and computer vision powered system for smart crop growth maximizing yield, efficiency and energy.

5) Multiplex_AI: An AI pipeline to process multiplex-FISH data to detect signal quality, non-specific signals, and signals in abnormal tissue types.

Publications

Banerjee TD, Raine J, Mathuru AS, Monteiro A. (2026) Modular fluidic automation platform with integrated thermal control for multi-step molecular imaging workflows. bioRxiv, https://doi.org/10.64898/2026.04.17.713973.

Goel T, Raine J, Kibat C, Collado JW, **Banerjee TD**, Mathuru AS (2026). Disruption of *chrna5* blunts aversion and adaptive transcriptomic responses to nicotine and alcohol. **iScience** 29, 114735. doi:10.1016/j.isci.2026.114735.

Banerjee TD[#], and Monteiro A[#] (2025). Shared regulatory networks link vein positioning and eyespot ring formation in butterflies. **Communications Biology.** ([#]Co-corresponding author). doi: 10.1038/s42003-025-09356-2. Featured article, Jan 2026.

Banerjee TD[#], Murugesan SN, Connahs H, and Monteiro A[#] (2023). Spatial and temporal regulation of Wnt signaling pathway members in the development of butterfly wing patterns. **Science Advances.** ([#]Co-corresponding author). Featured in the cover of 28th July 2023 issue. doi: 10.1126/sciadv.adg3877.

Tian, S, Lee, B, **Banerjee, TD**, Murugesan, SN, and Monteiro, A. (2025). A novel Hox gene promoter fuels the evolution of adaptive phenotypic plasticity Shen. **Nature Ecology and Evolution**. doi: 10.1038/s41559-025-02891-5.

Raine, J., Kibat, C., **Banerjee, TD**, Mathuru, A. S. and Biology, C. (2025). *chrna3* modulates biphasic response to alcohol. **Journal of Neuroscience**. doi: 10.1523/JNEUROSCI.0304-25.2025.

Banerjee, TD[#], Raine J, Mathuru, AS, and Chen, KH, Monteiro A[#] (2025). Spatial mRNA profiling using Rapid Amplified Multiplexed-FISH (RAM-FISH). ([#]Co-corresponding author) **bioRxiv**. doi: 10.1101/2024.12.06.627193.

Banerjee, TD[*#], Zhang, L[*]. and Monteiro, A[#]. (2025). Mapping Gene Expression in Whole Larval Brains of *Bicyclus anynana* Butterflies. **Methods Protoc.** 8, 2**.** **(***Co-first author and [#]Co-corresponding author). Doi: 10.3390/mps8020031. Featured in the cover of Volume 8, Issue 2.

Tian, S., Asano, Y., **Banerjee, TD,** Komata, S., Liang, J., Wee, Q., Lamb, A., Wang, Y., Murugesan, S. N., Fujiwara, H., et al. (2024). A microRNA is the effector gene of a classic evolutionary hotspot locus. **Science** (1141). 1141, 1135–1141. doi: 10.1126/science.adp7899.

Banerjee TD[*#], Finet C[*#], Seah KS, and Monteiro A[#] (2024). *Optix* regulates nanomorphology of butterfly scales primarily via its effects on pigmentation. (Special issue on structural colors in **Frontiers in Ecology and Evolution**; *Co-first author and [#]Co-corresponding author). doi: 10.3389/fevo.2024.1392050.

Prakash A., Dion E., **Banerjee TD**, and Monteiro A (2024). The molecular basis of scale development highlighted by a single-cell atlas of *Bicyclus anynana* butterfly pupal forewings. **Cell Reports.** doi: 10.1016/j.celrep.2022.111052.

How S*, **Banerjee TD*[#]**, and Monteiro A[#] (2023). *vermilion* and *cinnabar* are involved in ommochrome pigment biosynthesis in eyes but not wings of *Bicyclus anynana* butterflies. **Scientific Reports**, 1–12. (*Co-first author and [#]Co-corresponding author). doi: 10.1038/s41598-023-36491-9.

Prakash A, Finet C, **Banerjee, TD**, Saranathan V and Monteiro A. *Antennapedia* and *optix* regulate metallic silver wing scale development and cell shape in Bicyclus anynana butterflies Graphical. **Cell Reports**. (2022), doi: 10.1016/j.celrep.2022.111052.

Banerjee, TD[#], Tian, S., and Monteiro, A[#]. (2022). Laser Microdissection-Mediated Isolation of Butterfly Wing Tissue for Spatial Transcriptomics. **Methods Protocol**. ([#]Co-corresponding author). doi: 10.3390/mps5040067.

Wee J.L.Q., **Banerjee TD**., Prakash A., Seah K.S. and Monteiro A (2022). Distal-Less and Spalt Are Distal Organisers of Pierid Wing Patterns. **EvoDevo**, 13, 1–14, doi:10.1186/s13227-022-00197-2.

Banerjee TD[#] and Monteiro A[#] (2020). Molecular mechanisms underlying simplification of venation patterns in holometabolous insects. **Development.** ([#]Co-corresponding author). doi:10.1242/dev.196394.

Banerjee TD[#], Ramos D and Monteiro A[#] (2020) Expression of Multiple *engrailed* Family Genes in Eyespots of *Bicyclus anynana* Butterflies Does Not Implicate the Duplication Events in the Evolution of This Morphological Novelty. **Frontiers in Ecology and Evolution** 8:227. ([#]Co-corresponding author). doi: 10.3389/fevo.2020.00227.

Banerjee TD[#] and Monteiro A[#] (2020) Dissection of Larval and Pupal Wings of *Bicyclus anynana* Butterflies. **Methods & Protocol** 3, 5. ([#]Co-corresponding author). doi: 10.3390/mps3010005.

Connahs H, Tlili S, van Creij J, Loo TYJ, **Banerjee TD**, Saunders TE and Monteiro A (2019) Activation of butterfly eyespots by Distal-less is consistent with a reaction-diffusion process. **Development**, 146, 1–12. doi: 10.1242/dev.169367.

Banerjee TD# and Monteiro A# (2018) CRISPR-Cas9 Mediated Genome Editing in *Bicyclus anynana* Butterflies. **Methods & Protocol**. doi: 10.3390/mps1020016. (#Co-corresponding author).

Pal S, Kundu A, **Banerjee TD**, Mohapatra B, Roy A, Manna R, Sar P and Kazy SK (2017). Genome analysis of crude oil degrading *Franconibacter pulveris* strain DJ34 revealed its genetic basis for hydrocarbon degradation and survival in oil contaminated environment. **Genomics** 109, 374–382. doi: 10.1016/j.ygeno.2017.06.002

Pal S, **Banerjee TD**, Roy A, Sar P and Kazy SK (2015) Genome Sequence of Hydrocarbon-Degrading *Cronobacter sp.* Strain DJ34 Isolated from Crude Oil-Containing Sludge from the Duliajan. **Genome Announcement** ,3, 2011–2012. doi: 10.1128/genomeA.01321-15

Paul D, Kazy SK, **Banerjee TD**, Gupta AK, Pal T and Sar P (2015) Arsenic biotransformation and release by bacteria indigenous to arsenic contaminated groundwater. **Bioresource Technology** 188, 14–23. doi: 10.1016/j.biortech.2015.02.039

Reviewed papers in the following journals

eLife, BMC Biology, Annals of the Entomological Society of America, Cellular and Molecular Life Sciences, PloS genetics, Evolution and Development, Journal of Insect Physiology, Biology, Cells, Molecules, and Insect Sciences.

Conferences

1) 14th Zebrafish conference, Singapore, 2025. Spatially resolved mRNA profiling using Rapid Amplified Multiplexed-FISH (RAM-FISH)
2) Spatial Biology Congress, 2024, Singapore. Spatially resolved mRNA profiling using Automated Accelerated Amplification based multiplex-FISH (RAM-FISH).
3) Biology of Butterflies 2023, Prague, Czech Republic. Presentation. **Banerjee TD**, Murugesan SN, Connahs H, and Monteiro A. Spatial and temporal regulation of Wnt signaling pathway members in the development of butterfly wing patterns.
4) Euro-evo-devo 2022, Naples, Italy. Poster: **Banerjee TD** and Monteiro A: Reuse of a Wing Venation Gene-Regulatory Network in Patterning the Eyespot Rings of Butterflies.
5) 2nd Asia Evo-Devo 2021, Japan. Presentation: **Banerjee TD** and Monteiro A. Molecular mechanism underlying wing venation is reused to pattern the eyespot rings in *Bicyclus anynana* butterflies.
6) GYSS 2021, Singapore. Video presentation: **Banerjee TD** and Monteiro A. Molecular mechanisms underlying simplification of venation patterns in holometabolous insects.
7) Biology of Butterflies 2018, Bangalore. Poster: **Banerjee TD** and Monteiro A. Differential expression of *engrailed* paralogs elucidates gene duplication events in butterflies.

Teaching and mentorship

Current students

1) Research Assistant: Mr. Jeriel Lee Cheng Hock. **Project**: Investigating the spatial-temporal expression of leg gap genes during the embryonic and larval wing development of *Bicyclus anynana* butterflies.
2) FYP: Mr. Javen Tan Yih Ruay. **Project**: Modeling reaction-diffusion based mechanisms patterning the wing veins of *Bicyclus anynana*.

Past students

1) JayaVardhan Harihar Prasad and Chan Sum Ngai Jeff (Raffles Institution). **Project**: Design and implementation of nanolitre scale microinjector for CRISPR-Cas9 injection on embryos.
2) Gianella Pacho (UROPS). **Project**: Exploring FGF signaling in biological patterning using *Bicyclus anynana.*
3) Emily Zhang Linwan (Master Student). **Project**: Spatial transcriptomics of developing *Bicyclus anynana* brain.
4) Lara Chal (UROPS). **Project**: Co-localization of multiple eyespot centre-specific mRNA and protein targets in larval wings of *Bicyclus Anynana*.
5) Dorothy Lau Shi Min (FYP). **Project**: Involvement of EGFR signaling in biological patterning of butterfly wings.
6) Jeff Winxin Collado (FYP). **Project**: Implementation of automated fluidics system for in-situ hybridization experiments in the larval and adult brains of *Danio rerio*.
7) Teesha Basak (FYP). **Project**: Designing an optimized autofeeder for behavioral experimentation in Zebrafish.
8) Zhang Yu. Project (Master Student). **Project:** Spatial interaction of *frizzled* receptors in patterning butterfly wing tissues.
9) Jeriel Lee Cheng Hock(Amgen and UROPS summer student). **Projects:** i) Functional role of *optix* and *spalt* CREs in the development of venation and eyespot color patterns. ii) Spatial-temporal expression of leg gap genes during the embryonic and larval wing development of *Bicyclus anynana* butterflies.
10) How Hong Chuen Shaun (URPOS student) **Project:** Regulatory Interactions of *engrailed*, *optix* and Ommochrome Genes in *Bicyclus anynana* butterflies.
11) Tirunagari Praveenya (Summer Intern). **Project**: The role of Homothorax (Hth) and Dachshund (Dac) in the eyespot center formation in *Bicyclus anynana* butterflies.
12) Lim Jun Hui Solomon and Isaac Sim Jing Xiang (NUS High School). **Project**: Reaction diffusion modeling on wing patterns of butterflies.

Teaching Assistant duties: Five semesters at Department of Biological Sciences, NUS. Worked with Prof. Antónia Monteiro; Assoc. Prof. Ryan Chisholm; Asst. Prof. Roman Carrasco; Assoc. Prof. Li Diaqin; and Asst. Prof. Nalini Puniamoorthy.

Apps

https://tdblab.github.io/hcrprobedesigner/hcr_22.1.html
https://tdblab.github.io/rgbhcr/rgb_2.html
Merge: A python based tool to create composite 2D images of z-slices or maximum projection of *in-situ* data for multiplex-FISH applications. GitHub: https://github.com/tdblab/RAMFISH

Hobbies

Microelectronics, aquarium, cycling, computer hardware, guitar, gym, running, tennis, badminton, and photography.

Other Research/Industry Experience and Projects Undertaken

1) Summer Intern at the Lab of Microbiology, East India Pharmaceuticals Works Ltd., Durgapur Plant. Worked on detection and analysis of *S. abony, E. coli, P. aeruginosa,* and *S. aureus* in water samples.
2) Winter Research Intern 2013, Department of Biotechnology, Indian Institute of Technology – Kharagpur (2013-14). 'Utilization of Petroleum Hydrocarbon and Oxidation of As(III) by Bacterial Strains Isolated from Arsenic Contaminated Sites in West Bengal (India)'.
3) Summer Research Intern 2014, Department of Biological Sciences, National University of Singapore. Project title 'A Study on Expression of *engrailed* and *spalt* in Developing Wings of African Butterfly *Bicyclus anynana*'.

Other Participations and awards

1) Genomics, Metagenomics and Metabolic Engineering: Workshop organized by Indian Institute of Technology – Kharagpur, India'; December 2014.
2) EduHeal Foundation Interactive Olympiad (National Biotechnology Olympiad) 2009 and secured National Rank-389.
3) Advance Functional Material at CSIR-CMERI, Durgapur on 24th Jan 2013.
4) Won numerous awards in academics during school days.

Carrier Goal

Gain knowledge and apply it for the wellbeing of humanity.

Last updated on 12th May 2026